

The Local Stack

DeFi for Empowering Local Economies
“LoFi”

Financial Decentralized vs Economic Consolidation

Why build decentralized financial infrastructure if it's only going to be used to exchange goods with large, consolidated businesses and industries?

Decentralized finance should enable any economy, regardless of size, to become more resilient, and empower its members to shape how they interact amongst themselves and with the world.

To do this, “LoFi,” a subcategory of DeF for local finance, needs to have solutions that: 1) can be redeployed and customized, with minimal transaction and startup costs; 2) enable every person and entity to become a stakeholder; and 3) generate, circulate, and retain value in the local economy, and incentivize beneficial exchanges with other economies

This document is a starting point of “The Local Stack,” a term for sets of blueprints on the Radix decentralized ledger intended to create reusable financial infrastructure for any economy to use and shape the degrees of economic centralization or decentralization they desire..

The Problem: Economic Consolidation

There are two parts of this document where I highly suggest you read an outside article, and I think reading a bit of [BIG](#) by Matt Stoller is a much better primer for why local economies are needed. The control over the economy is getting consolidated in fewer and fewer hands, and having lots of negative consequences. Plus, our traditional tools for fighting back against this, like antitrust law, are failing us.

Next, there are other problems with modern value chains. Financial resources are generally extracted from local communities in exchange for mass goods and services, but there is not sufficient return in the bargain.

This structure also only exacerbates existing problems - for regions that have lost their key industries due to an evolving economy it's hard for them to develop new industries, in part because the broader economy's unfair bargain is undercutting many existing and potential enterprises.

The Problem: Economic Consolidation

Some of these problems may be even worse when looking at local economies on a global scale.

This document will not spend much time on diagnosing this problem, or other problems related to the success of local economies or the economy as a whole. Those are all valuable topics, but this document is intended for those who already believe that local economies matter. Similar to the 'Radix Revelation' it's just not worth diving into certain topics if someone hasn't already agreed with another, enabling viewpoint. I'm sure there will be plenty of discussion on these topics elsewhere, and will be happy to learn from them.

Also, I recognize that not all economic centralization is bad. For certain things there can be many benefits. Please don't take me as an absolutist - I just want communities to have more autonomy and agency.

The Solution: The Local Stack on Radix (1/3)

The goal of The Local Stack isn't to help a local economy. The goal of The Local Stack is to help all local economies.

To do this, the Local Stack needs to be built on RadixDLT, the first Layer-1 network purpose built for Decentralized Finance. The network satisfies three key pieces, in addition to general problems addressed by other networks like the “blockchain trilemma.”

First, the network must make it easy to redeploy applications for each economy. Radix achieves this with its blueprint method, which allows a new instance of an application to be instantiated on-ledger, meaning no code needs to be copied, changed, or created to create a new, secure and identical application.

The Solution: The Local Stack on Radix (1/3)

The second key piece required by The Local Stack is composability. Composability is the ability for applications and resources on a network to interact with each other, or even combine to create new applications. This is absolutely necessary for The Local Stack, as the stack is a collection of applications which all enhance local economic activity, and they stand to gain from being able to compound on each other rather than act as siloed institutions (like the status quo.)

Radix satisfies this as its consensus mechanism, Cerberus, doesn't only allow near infinite scalability, but the way its shards are braided together by nature creates atomic composability, meaning any application (or component as they're called on Radix) can directly interact with any other application or applications, and settle as one transaction.

The Solution: The Local Stack on Radix (1/3)

The final part of the infrastructure puzzle is scalability. The financialization of assets not only by The Local Stack, but by DeFi in general, plus existing financial transactions, means that the amount of transactions happening at any time will be orders of magnitude bigger than the current financial system. Further, segregating the financial world does not make sense - The Local Stack's goal is to enhance economies, not to isolate them. Financial exchange is still encouraged and should be as easy as possible - just with more autonomy. Additionally, it would be counterproductive for certain assets to not be linked across economies. For example, the metaverse is intended to create a new, shared world. Why should 2 local economies and the metaverse that connects them need to exist on three separate islands of financial infrastructure?

The Local Stack: First Steps

The Local Stack may eventually consist of dozens or hundreds of blueprints for different financial services. However, these may boil down to only a few services that are visible to the public, as many of the blueprints will be for discreet, specific pieces of functionality, or customizable options. Therefore, this document focuses on framing a handle of initial services The Local Stack can work towards creating (First Steps) but not necessarily all of the blueprints necessary to enable those services.

The First Steps discussed in this document are: 1) Securitizing rental units so renters earn equity, 2) crossing the bridge between The JOBS Act, and DAO's, 3) basic financial tools for small business DAO's, including payments, payroll, and inventory, 4) a local Complementary Currency System (CCS), and 5) an unbacked local Radix Universal Basic Income System (RUBIS).

Finally, it should be noted that these can be built by different organizations - there does not need to be a central entity to accomplish this agenda. Nor do they need to be built in any order. Further, hopefully this document and The Local Stack is a piece of inspiration for countless others to design LoFi solutions, as any contributions to local economies would benefit The Stack.

First Step: Securitized Rental Units (SRU)

The goal of SRU's are to: 1) give renters a chance to accumulate wealth, 2) incentive renters to act in the long term interest of the community. (Such as through other pieces of The Local Stack, or things like voting patterns.), 3) and obsolete substandard corporate landlords

SRU's do this by giving small pieces of equity to renters with each rent payment.

This document will exclude the specifics of how shares are accrued (such as time periods, size of each share, number of shares), and some mechanisms, not because they aren't important, but because they are variable, and all options may be on the final menu for each entity to pick.

The main part to convey about SRU's is that Radix enables the micro-securitization necessary to build this business model, and at the scale needed if The Local Stack movement goes global.

First Step: Securitized Rental Units (SRU)

There are pre-existing similar models, but more competition is not bad, and competitors can opt into any of The Local Stack blueprints as infrastructure for cheaper and broader service selections for their constituents. Also, with Radix's royalties system, this can be done in a way that brings rewards to the early builders who make fundamental infrastructure commonly used by other entities.

Building these models on Radix is ideal as the network offers a higher degree of customization - Eg adjacent traditional solutions often have conditions that limit equity earned to be used to purchase specific residences, or only provide benefits over a short time frame. Meanwhile Radix's "Radix Engine" and asset-oriented programming language, Scrypto, make it possible to design nearly any business arrangement that can be thought. For example, it would be practical and low cost to deliver a resident a fractional share of each rent payment for the next thirty years, based on equity accrued while a tenant, regardless of who owns the house. This means people may be more invested in the property, taking better care not just of the property, but also of the local community and its public and private institutions so they record more rent payments, as the value of the property and rent increase over time because of things like the resilient local economy they helped enable.

First Step: Securitized Rental Units (SRU)

Securitized Rental Units are a great entryway for LoFi conversations. Equity in housing is a known commodity. The Local Stack just proposes new ownership arrangements for that commodity. It's also a process that has a large use case. Plus it can have immediate benefits on both sides of the renter-landlord relationship, which could accelerate adoption.

The next steps for SRU's discussion would be to press The Local Stack's community to envision new arrangements that align the renter and landlord/financer/ownership DAO etc. This is a great thought exercise, and makes people start seeing how prioritizing local economies can matter. This could lead them to entering future conversations with an appreciation for local finance, and be in the lookout for compounding opportunities with more creative and innovative solutions elsewhere in The Local Stack or LoFi.

First Steps: The DAO-JOBS Act Spectrum

The DAO-JOBS Act spectrum is not a specific feature, but a recognition that local organizations, and especially businesses, will have certain regulatory obligations depending on their function and choices, and with great variance. In terms of current solutions for compliance there are things that are completely autonomous and permissionless, maybe something like a metaverse club, an entity that doesn't need any government regulation if it sticks to its purest form. Then, on the other side of the spectrum, is the closest thing currently allowed (by US securities law) to getting equity to supporters who aren't accredited investors - the ability to be part of crowd raising sites like StartEngine, which is enabled by The Jumpstart Our Small Businesses Act.

What The Local Stack aims to do about this spectrum is to build blueprints and organizations that help bridge the gap between these two options. For example, the JOBS Act is too cumbersome for a local entrepreneur to crowdfund from his local community to build something low scale but that's in local demand. Meanwhile, dispersing equity over a decentralized ledger with no legal incorporation whatsoever either isn't great or worth the risk for this type of project either. The Local Stack should aim to fill the gap between these options..

First Step: The DAO-JOBS Act Spectrum

The goal of exploring the middle ground of The DAO-JOBS Act Spectrum is to create methods by which community members can become stakeholders in local organizations and businesses. The definition of “stakeholder” should also be as broad as possible in this regard, encompassing both ways of contributing to or governing the organization, and/or ways to share in the value created by the organization.

This middle ground is extremely valuable as it gives community members a bigger reason to “shop local” rather than support the subset of unresponsive, centralized corporations that are offering a bad bargain for local economies right now.

First Step: The DAO-JOBS Act Spectrum

The types of entities that fill the DAO-JOBS Act spectrum will be numerous. Two U.S. states have already legalized DAOs as formal corporate entities, and more creativity will be needed to build a full set of offerings that combine the ability to execute a business plan with functions like crowdfunded equity or token governance.

The DAO-JOBS Act spectrum is also relevant for other parts of The Local Stack. For example, an apartment complex can be a legally incorporated DAO, and participate in the Securitized Rental Unit to give renters equity while also allowing both higher levels of investment or control via its DAO structure, or more traditional apparatuses depending on the specific goals of the group.

First Steps: The DAO-JOBS Act Spectrum

The DAO-JOBS Act Spectrum is fertile ground for expanding the LoFi conversation. The DAO topic in DeFi and crypto is growing, and LoFi can replicate ideas, such as structures, features, and use cases. Also, anyone can think of things they wish they had in their local community - it may even be reviving services that exited the local economy in the past for various reasons. Thus, people can look at a lot of DAO tools throughout DeFi, then ask: what type of stakeholder do I want to be with my local institutions, and which of these tools do I need to make that happen. Again, this could be a great place for more discussion about all the possibilities here.

First Step: Business Ops Tools

Business Operations Tools is simple: build tools to operate pieces of a business where relevant and practical. The goal of having these tools on the Radix network is to create more credibility for organizations that are doing things like crowdfunding.

For example, if a business ran its payment software on ledger it could send royalties directly to stakeholders, rather than just earning them a return on investment via increasing the value of the business and stakeholders' equity. This could prevent some fraud, as could payroll systems, inventory, affiliate programs, various FinTech solutions, and other services of executed on ledger. If seamless on-ledger solutions could build systems with a high level of trust then it could be an arrow in the quiver to justify different and more variable compliance requirements for legislation like The Jobs Act and future DAO laws.

First Steps: Business Ops Tools

Business Ops tools also have a direct selling point to businesses if they can be developed and used at lower cost than the fees extracted by traditional vendors.

Business Ops Tools are a great topic for LoFi. People understand what functions are necessary to run a business, and the use case for regulatory compliance gives an opportunity to not only show an answer to the “why on crypto?” question, but it also showcases that DeFi doesn’t always need to be about “secret actors” completely exiting the world and entering into a financial Wild West.

BOT’s should make it easy to design lots of services, then start sketching out solutions on Radix much faster than some of the other concepts which are building and designing new arrangements from the ground up and many permutations and designs will need to be vetted.

First Step: Complementary Currency Systems

Complementary Currencies are a subset of Alternative Finance which has existed since before Bitcoin.

The starting focus for The Local Stack in regards to CCS's is to develop systems similar to the BerkShares <https://www.berkshares.org/using-berkshares/> model, but on the Radix network so that it can easily be redeployed or customized. This makes it possible for any community to start a CCS and experiment with an infinite number of permutations.

The basic concept of the CCS is to build a token that is backed by another currency (Eg how BerkShares are backed 1:1 with USD deposit in local banks), but restrict or program how it can be used so as to promote and prioritize local economic growth.

First Steps: Complementary Currency Systems

The simplest way to build a CCS on Radix would likely involve wrapping another token, Eg XRD, or USDC (or similar token people will likely develop for Radix), then having special features for the wrapped token. This is distinct from creating a token whose value is based on any inherent value of the local economy - the CCS doesn't need to build up its value - it has the value of the tokens that are backing it, but enables new types of social compacts to be made and for participants know exchange it for any type of transaction.

The special features can be endless. For example, it can operate like BerkShares, in which only certain people may use it (for BerkShares digital program), and there is an exit fee, or it can be feature new innovations like a "tax" levied by transacting parties if they don't have the appropriate "local" Badge (Badges are a resource type on the Radix network). This means the wrapped tokens can be used with any counterparty, but are most beneficial and cost effective if you keep it within the community of badge holders. Complementary Currencies make it easy to organize collective action to prioritize local in various fashions. Things like taxes can also be used in innovative ways, such as funding a DAO, or direct payments to all Badge holders.

First Step: Complementary Currency Systems

Complementary Currency Systems are a really really fun thing for hypothetical conversations about LoFi. While they have a history, it's something that's not widely used in modern society. Thus, when you really take the time to read about what they are (I highly recommend reading about BerkShares for a working example with concise explanations), and really dive into Radix to understand it can be used to program almost any financial arrangement, even for the currencies themselves, and understand what it means to instantiate replicas of them on ledger, then you have a completely blank canvas to start designing your own CCS or CCS tools, without too many preconceived notions as I'm guessing most readers aren't familiar with Complementary Currencies. If you don't want to think of your own designs, then step out of the crypto subcategory of the Alternative Finance movement, and look at various systems that have been built with good ol' paper, then just imagine digitizing it.

First Steps: Complementary Currency Systems

Extra nuggets, an example of possible CCS in action: Imagine a token, free to transact between holders of a specific badge, 1% fee if traded between a badge holder and non badge holder, 3 percent fee to transact between both non badge holders, or 5% fee to just unwrap the token so it's not locked into the CSS anymore.

In this system grocery stores may have an incentive to stock more products from local producer & preparers, rather than just mass produced food, as the local vendors could be an outlet for the grocery store to spend some of its CCS tokens taken in from customers. Meanwhile, when making the purchase, the customer doesn't need to fret about finding local products in store - there a general agreement and economic incentive for the store to be as local as possible, but know if the nitpicking over details about suppliers and practices and geography etc.

This is actually just an accelerant to existing trends and further localizes them. Grocers are already adding items from smaller companies so that they have unique options to distinguish themselves from other mass product retailers, which in suit has caused major corporations to start diversifying their products, or offering special items to specific retailers. Also, with the rise of better, cheaper, and automated last mile solutions it may not be far in the future where local logistics networks could tie in a variety of vendors that together can displace a mass market grocery store altogether. Added bonus: the local vendors can exist on the DAO-JOBS Act Spectrum, enabling extra benefits for community members, and the community, if they are partial to participation.

First Steps: Radix Universal Basic Income Systems

The Radix Universal Basic Income Systems (RUBIS), is likely the simplest idea in this document to implement, but most ambiguous about what type of value it will provide.

RUBIS are not backed by any value (as CCS is) - they are just a unique coin (specific to the local economy) on the Radix ledger that has a specific amount minted each month and doled out to all accounts that have the requisite Badge. That's it.

The goal is to introduce a new unit of account for people to use to encourage utilization of underused or overlooked community resources and people, and to open up benefits which actors may want to bestow on community members. Eventually it could turn into a form of social currency, but it can take many directions and you can hopefully get a vague idea about all the possibilities as you've incrementally moved through these increasingly innovative ideas for LoFi.

....Okay, it's decentralized “synergy” for local economies

Yes, it has been a buzzword over the years, but hopefully what this document covers lead you to 1) understand the importance of local economies (there are many other issues with the economy though, so do not take this as a full solution, or full discussion of the economy in general or all local economies); then 2) showed you sample types of innovative financial tools that can be built on the Radix network and some explanation of why Radix is key for scaling this specific type of DeFi subcategory because of things like easy replication; and 3) finally with great hope that it causes you to continue thinking of better ideas (In me vs. the world I bet on the field to come up with a better idea in this domain.), and understand that all of these “Local Stack” tools for “LoFi” compound ... or synergize with each other, and are a ripe area for new innovative in the DeFi space, or at least some fresh conversation.

Appendix

Pre-existing Models: Securitized Rental Units

Up&Up:

“After that, every month, that share of the property increases as they pay their rent. In addition, if the house appreciates in value during the course of the lease, then the renter earns more. And should renters want to do so, they can convert their ownership stake into a down payment to buy either the home they’re living in or another Up&Up property. At the end of the lease, they have the option of cashing out completely.” [[Forbes](#)]

SRU Unique Selling Points: 1) More flexibility in vesting schemes, royalty schemes etc. - can create more long term concern about the community - e.g. “what if I’m my earnings from this property aren’t just it’s change in value over next 3 years, but next 30?” Most important thing is that this flexibility can be endless, and scaled globally, rather than a specific set of offerings from 1 company; 2) Competition here is fine. Anything to get renters more wealth and to have them focus on community development then the better off we are. Also, we should be agnostic - in an ideal world they could just start using our blueprints and offer their customers more options. 3) composability and interoperability with the rest of The Local Stack, and other housing organization on The Local Stake [e.g. like non-profit version of Up&Up, or the Renter Equity Program on next slide, 4) all assets will move to DeFi eventually, might as well skip an intermediary piece of technology.

Pre-existing Models: Securitized Rental Units

[Rental Equity Programs](#)

Future of this deck.

I don't know how or if i'll update/edit this deck, but I really hope you enjoyed it. If you want to talk about any ideas then just come hang out in the growing radix ecosystem and channels. I'm definitely interested in talking with the community and working through ideas, and if you got this far I'm guessing you do to.

Also, I know there can be lots of unintended consequences for things like this. I originally wanted to do a 5 slide deck and it ended up almost thirty and I realized I had to exclude a lot of discussion. But, we're living in a decentralized world! I genuinely think this idea will flourish if discussed in public, and DeFi is a fast moving space so early collaboration on any ideas, even if it's to eliminate them as an option, is valuable.

Funny thing about “decks”

I think that last slide was the first time I used the word “deck” in this document, and it actually led me to remember a funny thing about Radix and spreading the word with friends.

I found it's really fun to make your own meme coins, then show your friends that you created your own crypto. I've created OnionCoin, and DeckCoin as meme coins, with no purpose and no value. It definitely solicited some of the most interest I've seen from several friends.

So, Mentioning “decks” reminded me of DeckCoin, so here's its story: A friend was planning to get his deck replaced last year. Then a tornado hit. Not only did that ruin #DeckPlans, but caused major damage to the house. Long story short, the house is finally back in order, and the deck is getting ready for inspection! So to celebrate, I created DeckCoin for his deck. The first deck with its own currency.

Hit me up if you want any of this pointless coin (free), and share it with your friends on any deck related activities you want, whether it's reading powerpoints, playing poker, on soaking in some sun.